Introduction to Life Cycle Assessment (LCA)

CIVE475 Summer 2025 Life Cycle Assessment

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Lecture objectives

- Introduce the concept of product life cycle assessment (LCA)
- Define LCA, its components and distinguishing factors
- Understand uses and users of LCA

LCA

Life Cycle of a Product

Cradle
raw material extraction

Cradle-to-cradle: *aims to create production techniques that are waste free*

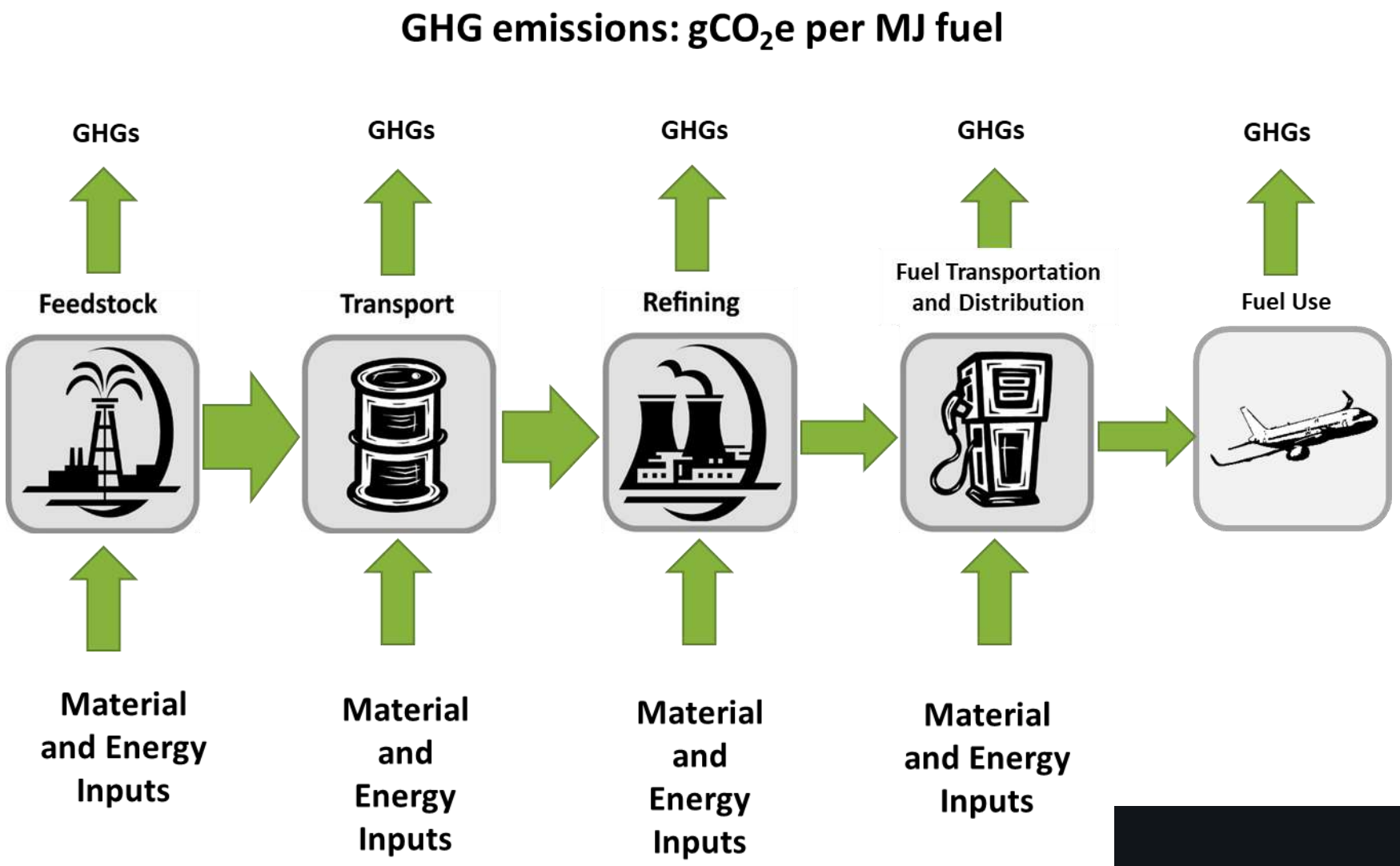
Cradle-to-gate: *resource extraction through product production/distribution*

Grave
product end-of-life

Gate
product exits production facility

Cradle-to-grave: *resource extraction through product end-of-life*

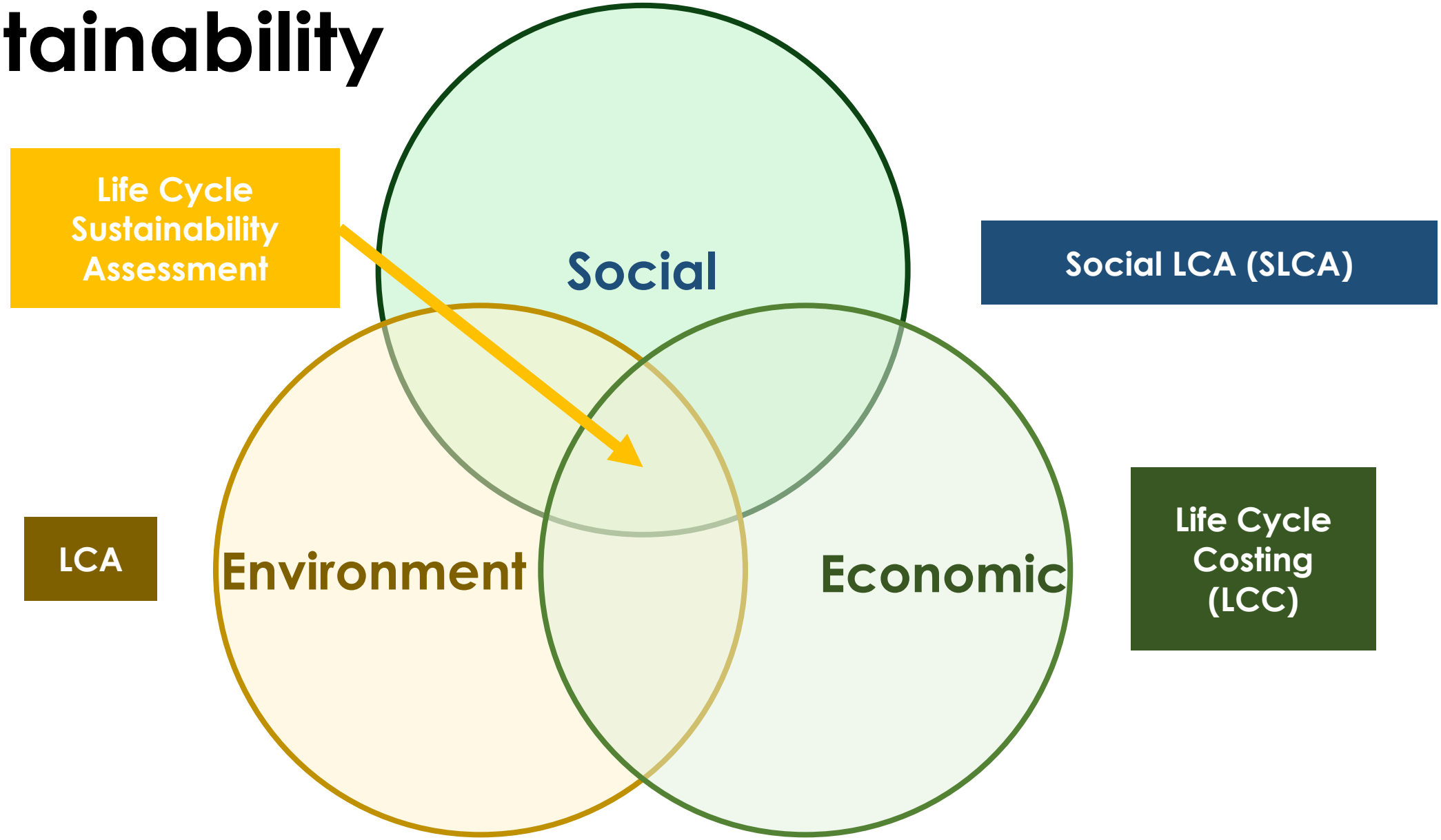
Example: Life Cycle of Jet Fuel



Life Cycle Assessment (LCA)

- LCA is a **systematic** tool to quantify potential environmental impacts over the life cycle of a product or a service
- Environmental dimension of life cycle thinking
- **Captures a comprehensive chain of events and activities** that are necessary to support the product or process
- **Extends** the "normal" analysis of looking at the obvious
- Relative approach based on the **functionality** of a product/service
- **Does not predict** absolute or precise environmental impacts

LCA and the Three Pillars of Sustainability



Guidance for Conducting an LCA

LCA is a standardized framework

ISO 14040: LCA Principles and Framework

ISO 14044: LCA requirements and guidelines

Why?

- Provides guidelines for practitioners
- Ensures consistency and credibility
- Enhances comparability of results
- Reduces potential for manipulation

BUT: "there is no single method for conducting LCA" (ISO 14040)

Types of LCA

Screening LCA

- Follows the ISO 14040 / 14044 framework
- No critical review needed
- No reporting requirements
- Internal purposes only

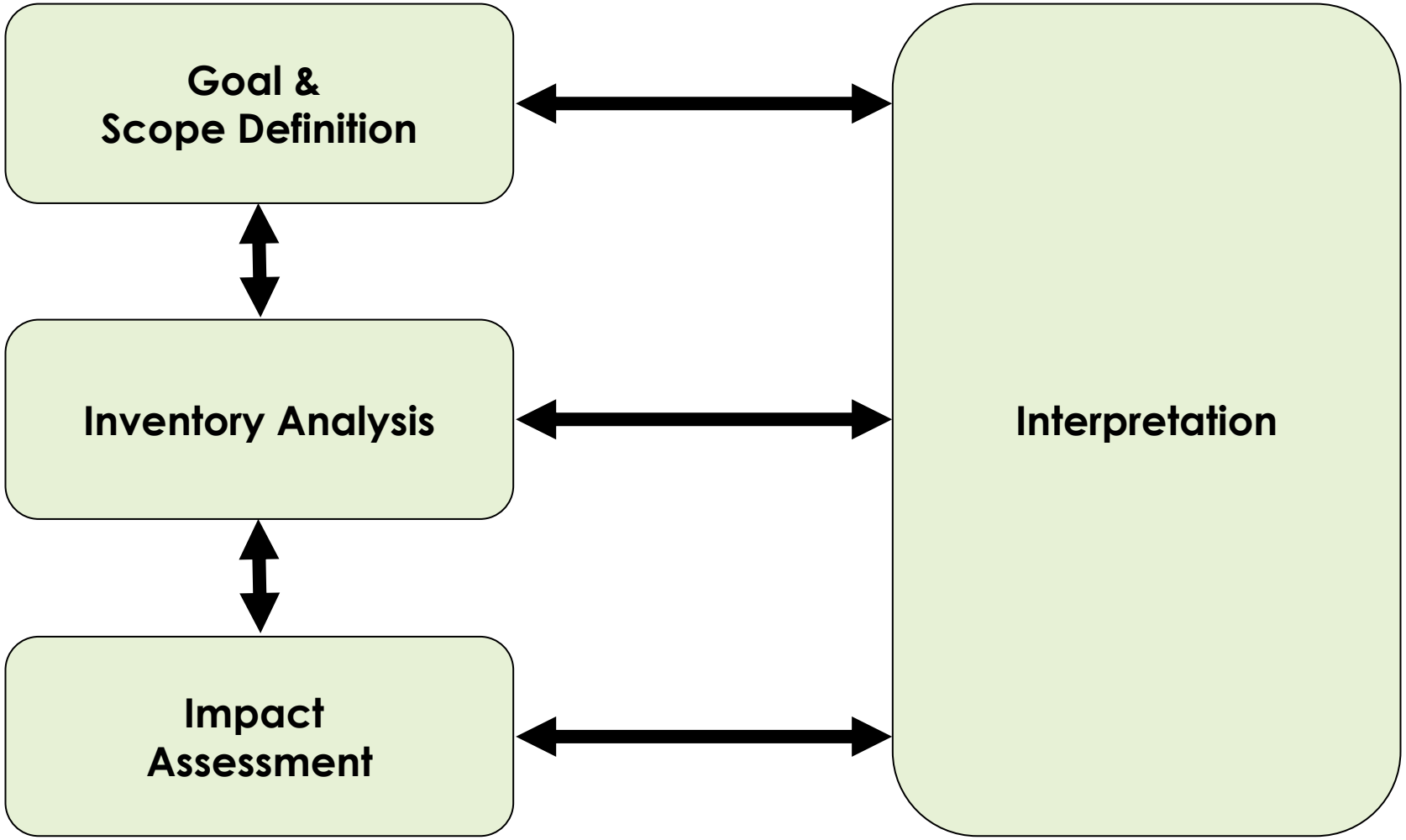
Single product detailed LCA

- Reporting requirements
- One single independent reviewer
- Third-party report for external parties
- Can be used for environmental claims

Comparative detailed LCA

- More extensive assessment and reporting requirements
- a panel of at least 3 external reviewers
- Third-party report
- Can be used for environmental claims

LCA Framework



Other ISO Standards Connected to LCA

- ISO 14067 on the carbon footprint of products
- ISO 14020, 14021, 14024, 14025, and 14026 on environmental labels
- ISO 22526 on carbon footprint and removals for biobased plastics
- ISO 20915 on life cycle inventory for steel products
- ISO 14072 on effective application of LCA to organizations
- ISO 14046 on water footprint
- ISO 14063 on environnemental communication
- ISO 14080 on greenhouse gas management and climate actions

LCA Applications

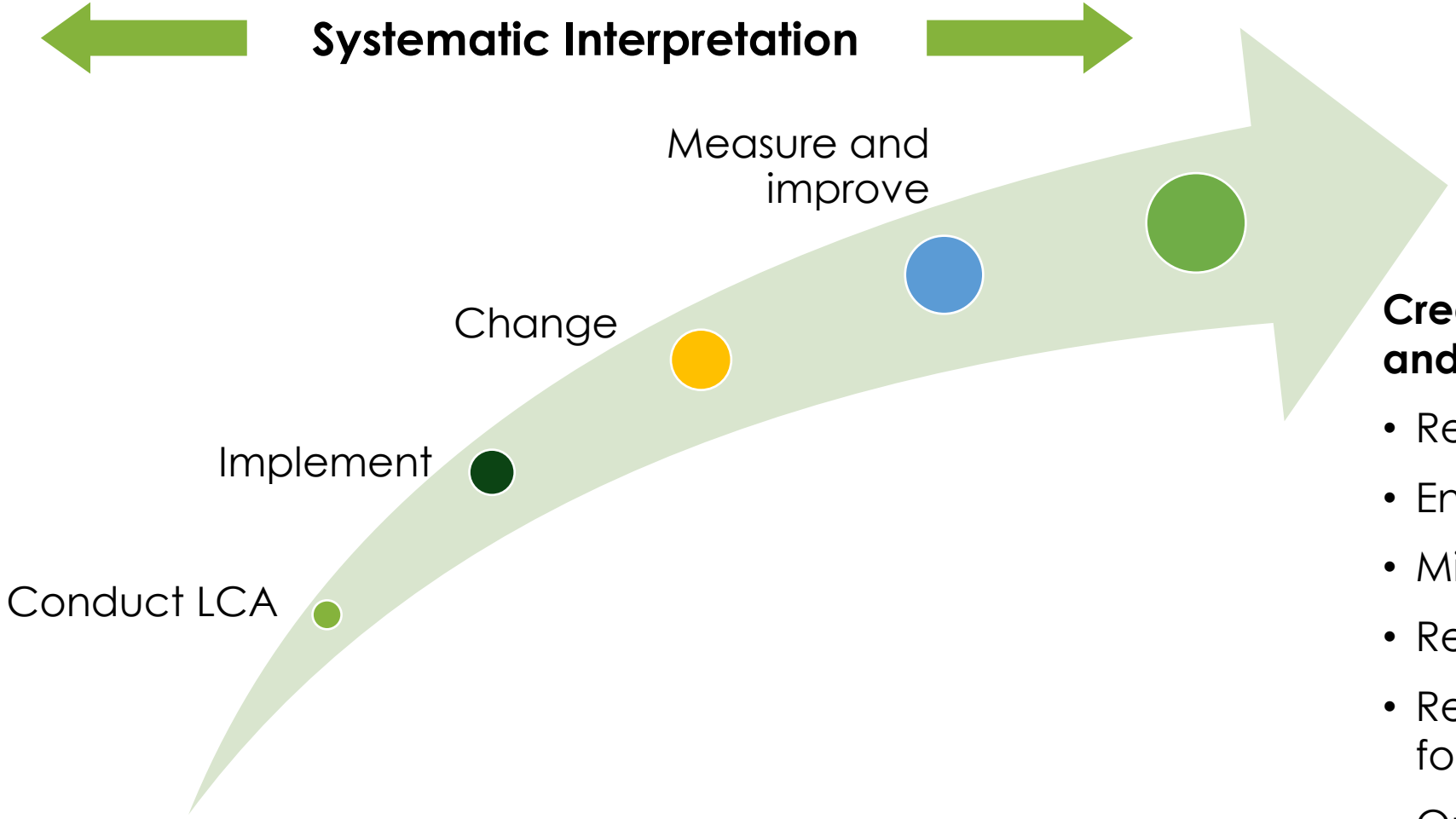
Practice Problem

A clothing store hires you as a consultant to conduct a comparative LCA of plastic vs. wood hangers. Provide two possible reasons for which the store would want to conduct this study.

Uses of LCA

- **Evaluate** the impacts of a product system
- **Identify** the largest sources of impacts
- **Compare** the impacts of competing products systems
- Help **manufacturers** improve their processes
- Help **consumers** make informed decisions
- Help **policy makers** anticipate risks in policy interventions
- Help **researchers** set priorities for product improvement
- **Inform** strategic sustainability planning
- **Ensure** accurate marketing and environmental claims
- **Inform** funding agencies & investors
- **Enforce** policy

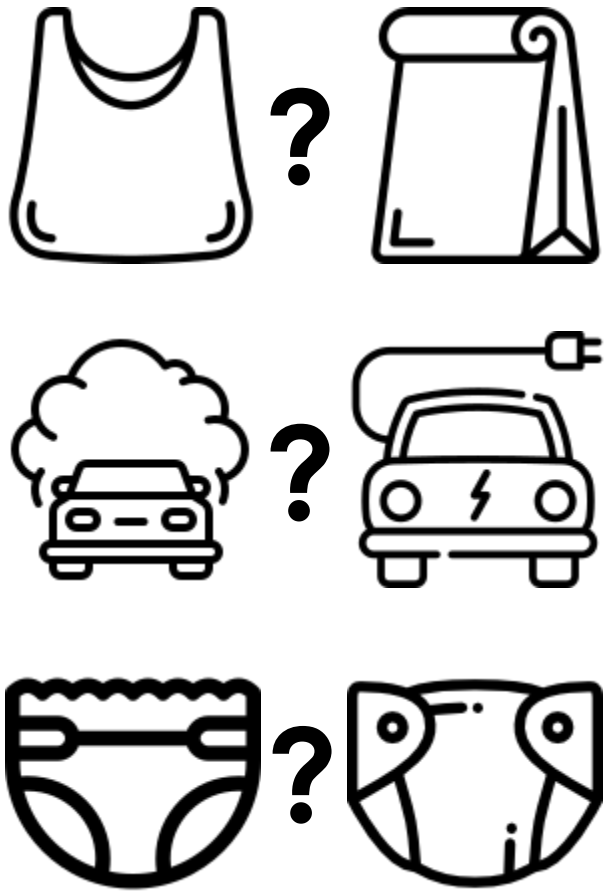
Focus on Impact Not Study Results



Create positive business value and sustainability impact

- Revenue growth
- Enhanced brand
- Mitigated risks
- Reduced cost
- Reduced environmental footprint
- Optimized used of resources
- Etc.

Use of LCA: Comparative analysis



Objectives:

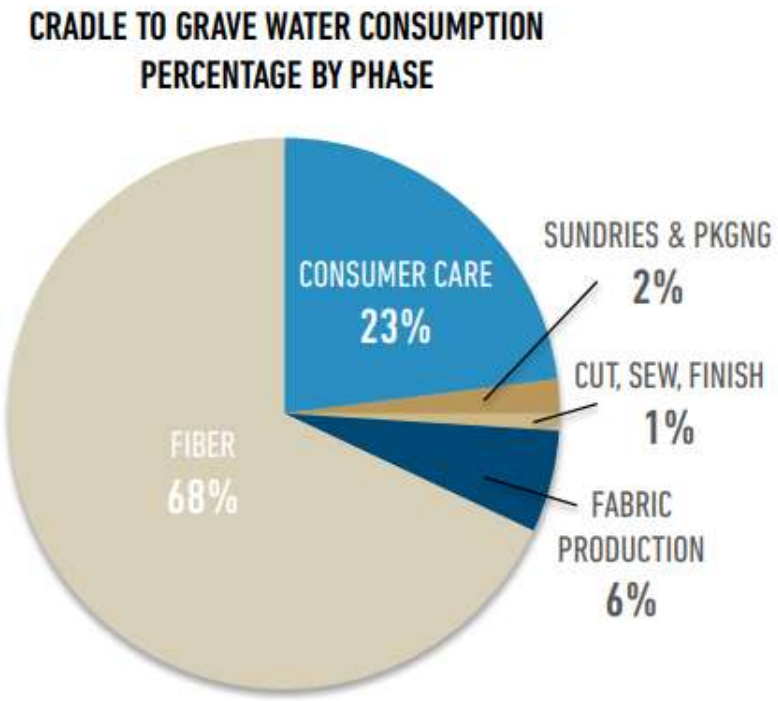
- Help make informed decisions on suppliers / input materials
- Quantify sustainability implications of choices
- Reduce environmental impacts of activities/products/services

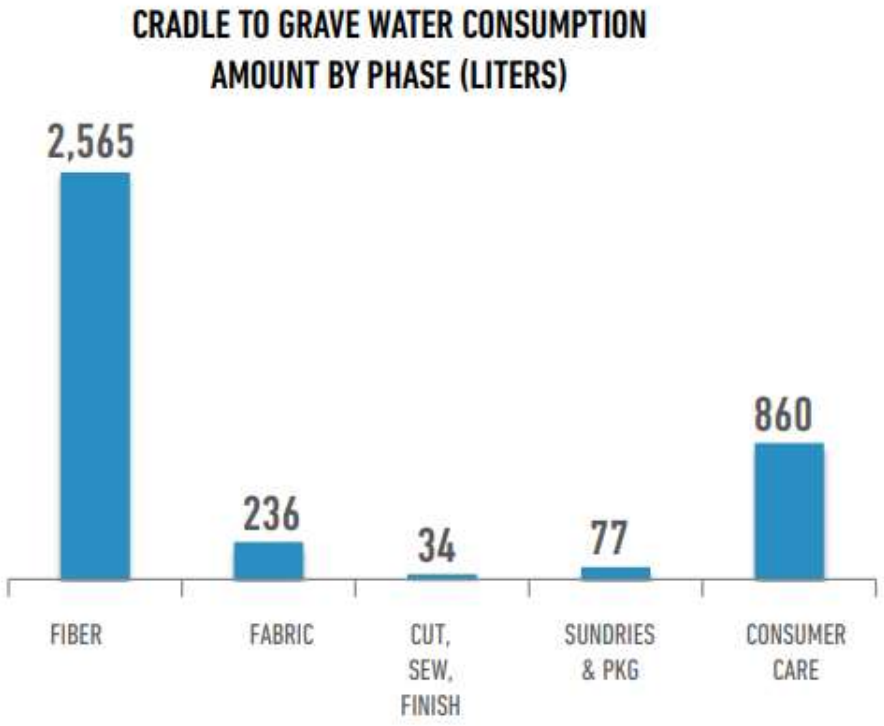
Use of LCA: Market differentiation

Use of LCA: 'Hotspot' / contribution analysis

Objective: Identify the life cycle stages that contribute most to the environmental impacts

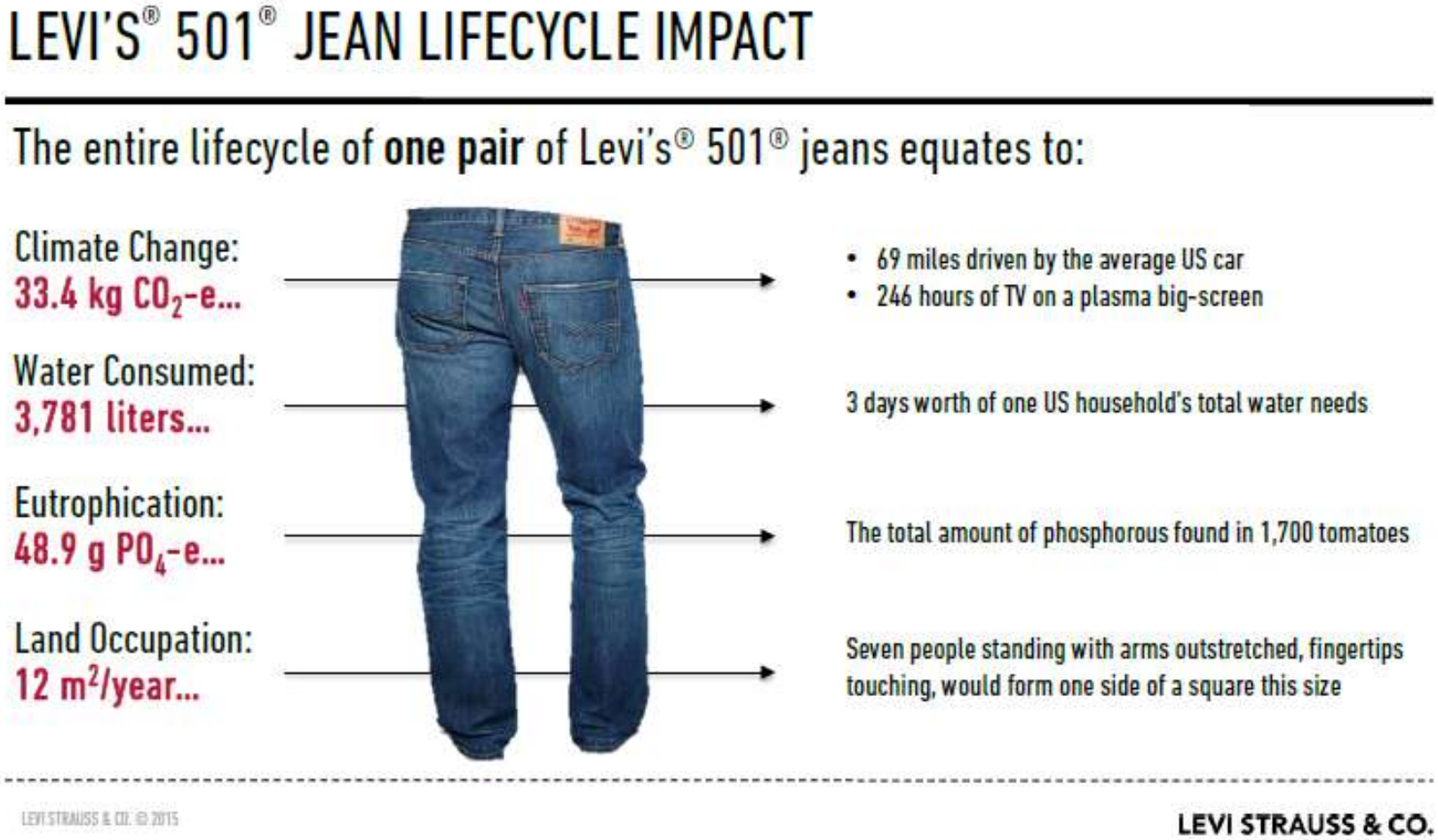
LCA of a pair of jeans



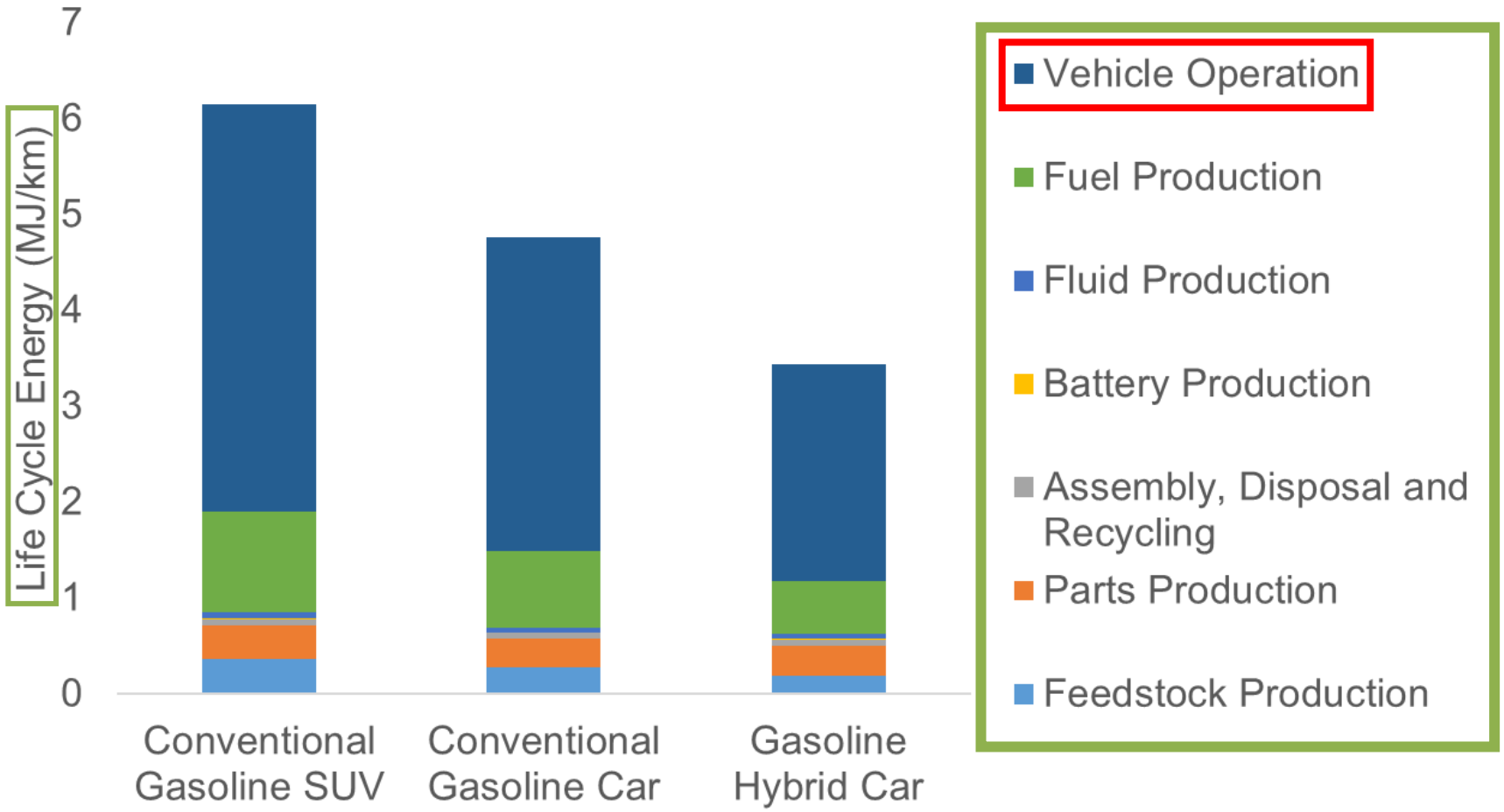


Source: Levi Strauss & Co

Use of LCA: Consumer engagement

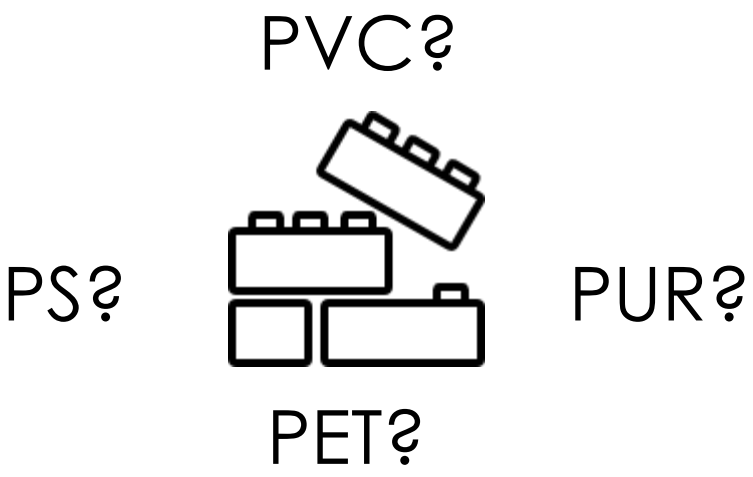


Use of LCA: Comparative and contribution analysis



Source: Argonne GREET Model

Use of LCA: Eco-design



Objectives:

- Help prioritize potential mitigation solutions
- Provide a comprehensive picture (from cradle-to-crave) of design choices to potentially limit environmental burdens of end-of-life
- Measure sustainability implications of design choices

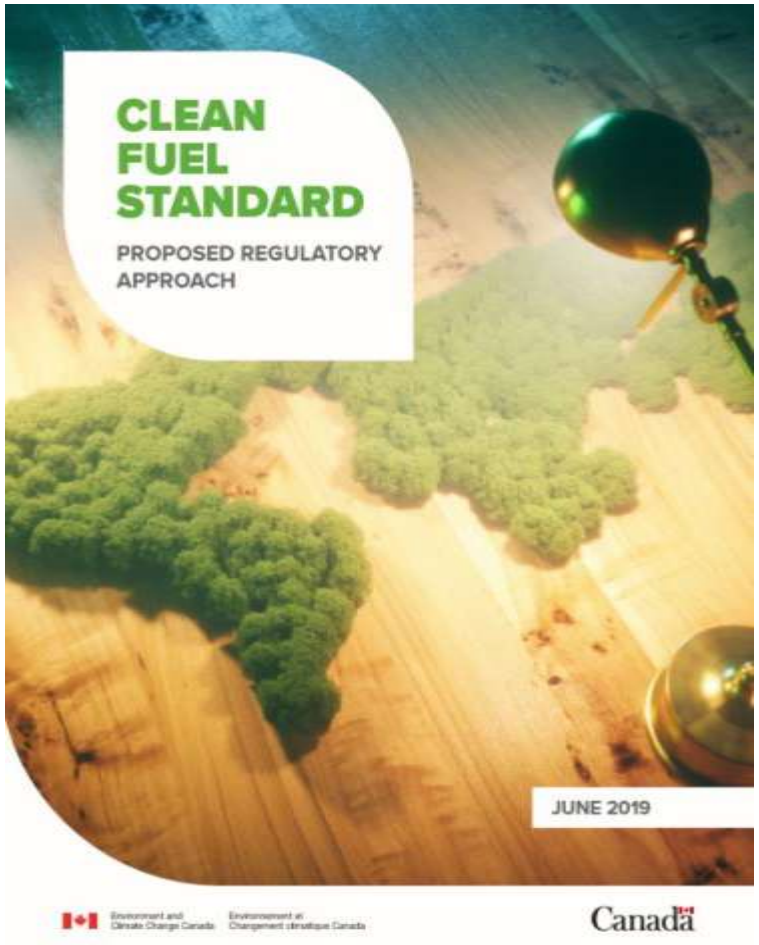
Use of LCA: Product innovation

Case study: After commissioning an LCA of detergents and determining energy to heat water was one of the main hotspots, P&G created a new detergent designed to work with cold water.

Environmental Product Declaration (EPD)

- Standardized (ISO 14025) document about a product's potential environmental impact
- Short version of an LCA
- Normally conducted by the product manufacturer and verified by independent expert
- Cover a large set of product categories (e.g., from chemical to construction products)
- Used for: Providing verifiable and accurate information on product's footprint, green public procurement, building assessment tools (LEED, BREEAM, GreenStar)

Use of LCA: Policies



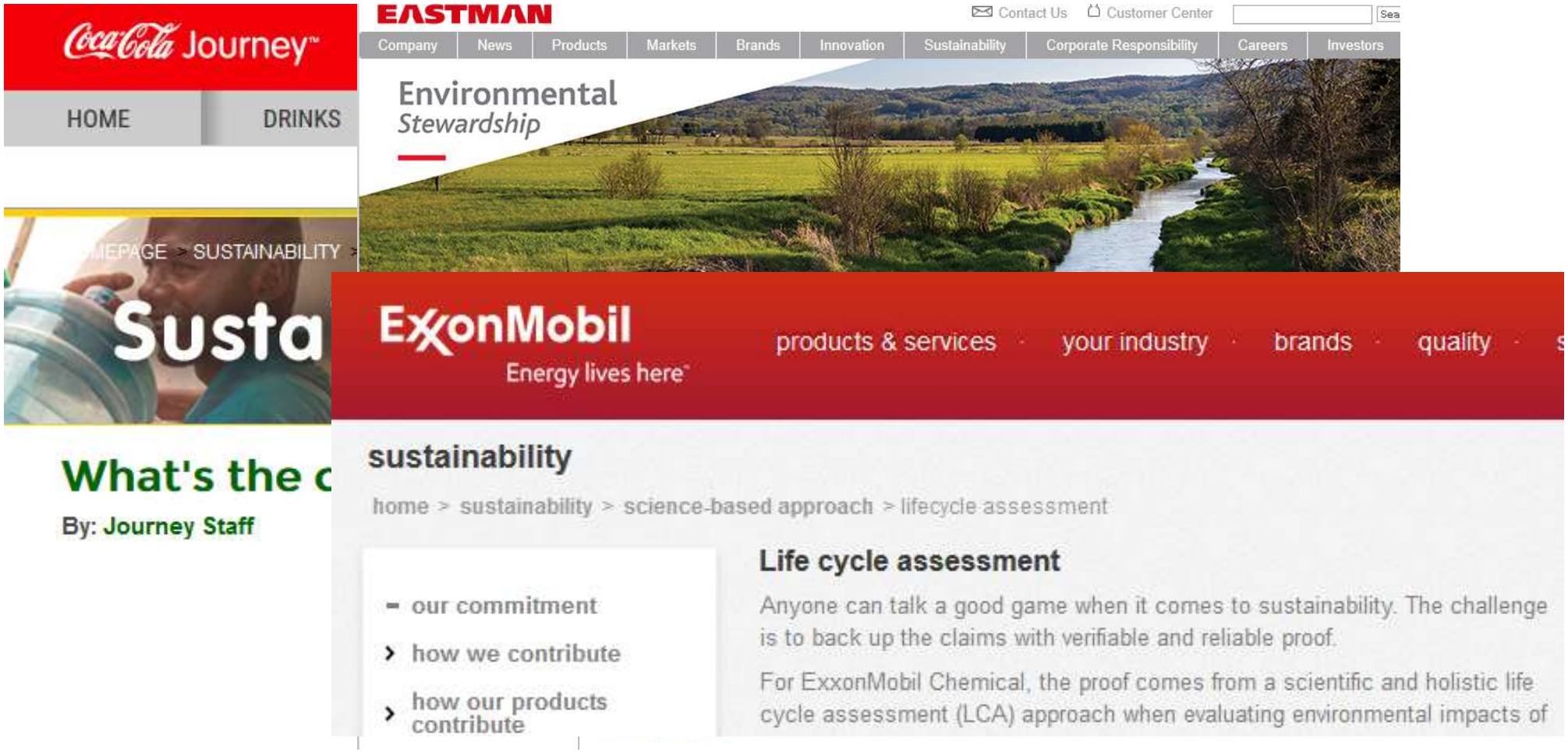
Objectives:

- Help to set indicators and targets
- Help to enforce the targets

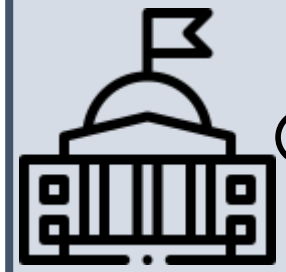
Example: the Clean Fuel Standard in Canada aims to reduce life cycle carbon intensity of fuels and energy used in Canada

Source: Environment and Climate Change Canada

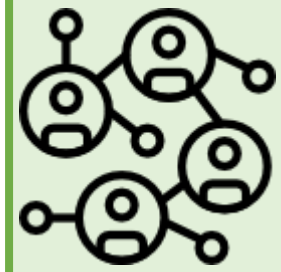
Use of LCA: Corporations



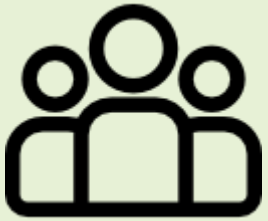
Audience for LCA



Government



Industry associations



Citizens

Academia

Corporations

Engineers

Limitations of LCA

- A model with large **uncertainties**
- **Contextual** and not generalizable
- Estimates how "**efficient**" a product/process/service is over its life cycle and does not determine if it is sustainable
- Assumes **linearity**
- The use of an LCA matters **more** than the LCA itself
- **Counterproductive** implications can happen from misuse of an LCA
- **Gaps** in toxicity impact models
- Many chemicals are **not characterized** and evaluated in LCAs

What have we learned?

- Concepts of life cycle thinking, life cycle of a product and LCA
- History, uses and users of LCA
- Unique attributes of LCA